

Math 110 – Homework 1 Answers

Math 110 – Fall 2026

Homework 1

Sets, fields, vector spaces, and subspaces

Suggested completion date: Friday, September 4

Unless otherwise indicated, all vector spaces are over the field $\mathbb{F}$. Problems referring to LADR are from *Linear Algebra Done Right*, 4th ed..

Problem 1. Let $f : X \rightarrow Y$ be a function. For $U \subseteq X$ and $V \subseteq Y$, define

$$f(U) = \{f(x) : x \in U\}, \quad f^{-1}(V) = \{x \in X : f(x) \in V\}.$$

- (a) Prove that $f(f^{-1}(V)) \subseteq V$.
- (b) Prove that $U \subseteq f^{-1}(f(U))$.
- (c) Give examples showing that either inclusion can be strict. Then determine exactly when equality holds in each of (a) and (b).

Problem 2. LADR §1A, Exercise 7.

Problem 3. Let $a, b \in \mathbb{F}$. Prove that if $ab = 0$, then $a = 0$ or $b = 0$.

Problem 4. LADR §1B, Exercise 2: if $a \in \mathbb{F}$ and v is a vector such that $av = 0$, prove that $a = 0$ or $v = 0$.

Problem 5. LADR §1C, Exercise 4.

Problem 6. LADR §1C, Exercise 5.

Problem 7. Work over the field $\mathbb{F}_2 = \mathbb{Z}/2\mathbb{Z}$. Expand $(x+1)^8$ by first computing $(x+1)^2$ and then repeatedly squaring. Explain where the calculation differs from the corresponding calculation over $\mathbb{R}$.

Solutions

Problem 1

Solution notes

Problem 2

Solution notes

Problem 3

Solution notes

Problem 4

Solution notes

Problem 5

Solution notes

Problem 6

Solution notes

Problem 7

Solution notes
